Mustafa Muhammad Alhasson

Software Engineer

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SUMMARY

Software engineering graduate (B.Sc. 2026) building full-stack products with TypeScript, Next.js, and Node.js — most recently a RAG-based document Q&A service with an async ingestion pipeline, hybrid vector retrieval, and an automated evaluation harness. Shipped production features in logistics automation and institutional web platforms; seeking a junior software engineering role.

EXPERIENCE

JavaScript Developer — Mundoimex Global Lojistik Mar – May 2025 · Istanbul, Türkiye

- Built and deployed an automated price-inquiry script that replaced manual data entry in logistics quoting workflows, cutting request processing time.
- Analyzed operational shipment data to verify the accuracy of algorithmic pricing calculations.

Web Developer & Researcher — IAU UNESCO Oct 2024 – Jan 2025 · Istanbul, Türkiye

- Designed and implemented modular web pages for the official IAU UNESCO platform, improving interface consistency across the site.
- Diagnosed and fixed site vulnerabilities and bugs, improving platform stability and load performance.
- Delivered concurrent technical tasks on tight schedules in a small cross-functional team.

PROJECTS

DocQA — RAG document Q&A service — *TypeScript, Fastify, Next.js, PostgreSQL (pgvector), Redis, BullMQ, OpenAI API* github.com/pascalmmk/docqa

- Built an end-to-end RAG backend: async ingestion pipeline (parse → chunk → embed) with BullMQ retries, dead-letter queue, and checkpoint resume; hybrid retrieval fusing pgvector similarity and Postgres full-text search with reciprocal rank fusion.
- Streamed answers over SSE with inline citations mapped to source chunks; protected the LLM budget with per-user token-bucket rate limiting and layered Redis caching.
- Wrote an automated evaluation harness (30 hand-written Q/A pairs) that gates CI on retrieval quality — currently 100% hit@8 — plus LLM-as-judge answer-faithfulness scoring; deployed on Fly.io, Vercel, and Neon.

EcoTrack — carbon & ESG reporting platform — *Next.js, TypeScript* app.ecotrack.dhsyazilim.com

- Built and deployed a live dashboard for industrial clients to monitor, calculate, and report greenhouse-gas and ESG metrics.
- Designed granular data-ingestion pipelines that keep emission sources separate (e.g., diesel vs. natural gas) to meet compliance reporting standards.

Coup — real-time multiplayer board game — *Flutter, Dart, Riverpod, Firebase* github.com/pascalmmk/coup_game

- Built a cross-platform (web, mobile, desktop) multiplayer adaptation of Coup with complete game logic: actions, character abilities, challenges, and blocking.
- Used Cloud Firestore and Firebase Auth for low-latency session-state synchronization and player authentication across concurrent games.

Soundcharts music analytics backend — *Node.js*

- Engineered a backend service integrating the Soundcharts API to track real-time music-industry charts, with rate-limited low-latency endpoints, structured error handling, and automated background data retrieval.

EDUCATION

B.Sc. Software Engineering — İstanbul Aydın University 2022 – 2026 · Istanbul, Türkiye

American High School Diploma — Elite Academy International School 2019 – 2021 · Istanbul, Türkiye

- GPA 3.45/4.0 · SAT: 1100 (Math 580, Evidence-Based Reading & Writing 520).

SKILLS

Languages: TypeScript, JavaScript, Python, C#, Java, Rust, Dart

Frameworks & tools: Next.js, React, Node.js, Fastify, PostgreSQL (pgvector), Redis, BullMQ, Docker, GitHub Actions, ASP.NET, Flutter, Tailwind CSS, Firebase, Git

Machine learning: Neural-network training & optimization · Udemy: Deep Learning A-Z, Machine Learning A-Z

Spoken languages: Arabic (native), English (C2), Turkish (B1)